

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460269.6187 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.164 +/- 0.017
Transit depth (Rp/Rs)^2	2.68 +/- 0.56 [%]
Orbital Inclination (inc)	84.91 +/- 1.18
Airmass coefficient 1 (a1)	1.153 +/- 0.014
Airmass coefficient 2 (a2)	-0.124 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	None
Optimal Aperture	3.83
Optimal Annulus	7.94
Transit Duration (day)	0.0721 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.164 ± 0.017 vs 0.1646 (deviation 0.04σ)
Duration fit vs published	0.0721 d vs 0.0758 d (deviation 0.45σ)
Mid-time O–C	-4.4 min
Depth SNR	9.6
Residual scatter	1.26 %

Light curve

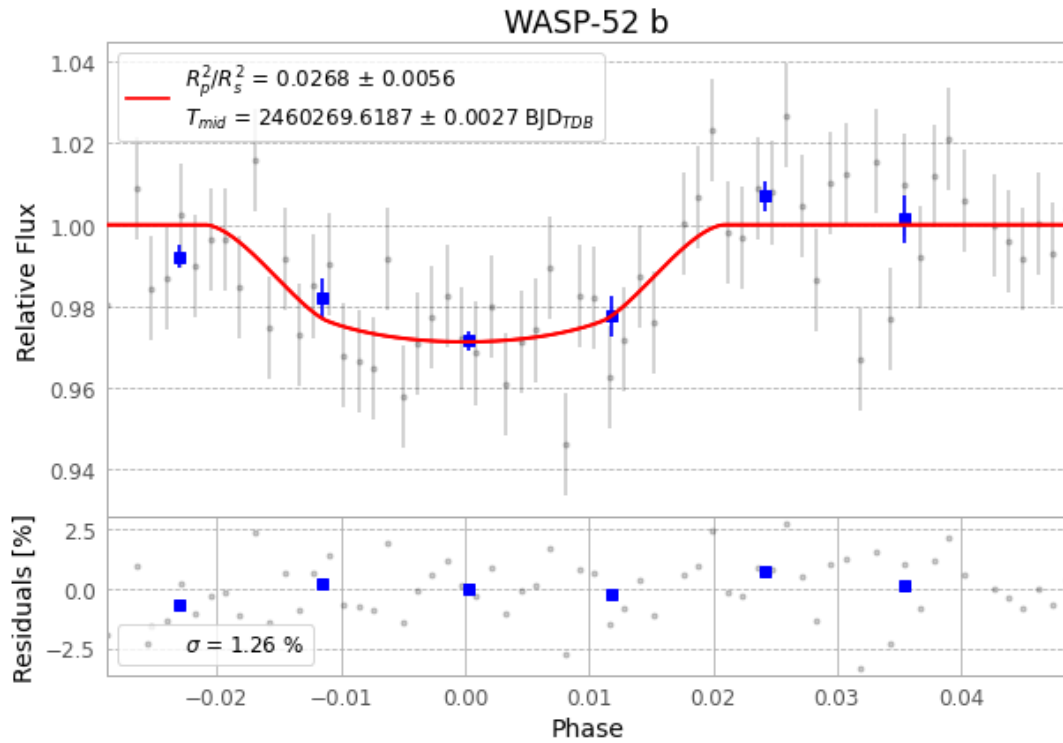


Figure 1 — FinalLightCurve_WASP-52 b_2023-11-20.png (SHA-256 e08887891c060557...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [313, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9eefb7b793018c8a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

65 FITS frames (43 MB), WASP-52231121013314.FITS → WASP-52231121044815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-231121.log (fd9eb2189b0b...), FinalLightCurve_WASP-52 b_2023-11-20.png (e08887891c06...), FinalLightCurve_WASP-52 b_2023-11-20.csv (144b4a0b0157...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 22:01Z.